BRAC UNIVERSITY

CSE 350: Digital Electronics and Pulse techniques

Exp-04: Analysis of the binary weighted D/A converter

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Objectives

1. To construct binary weighted D/A converter
2. Verifying that the digital signal is converted to a proportional analog signal

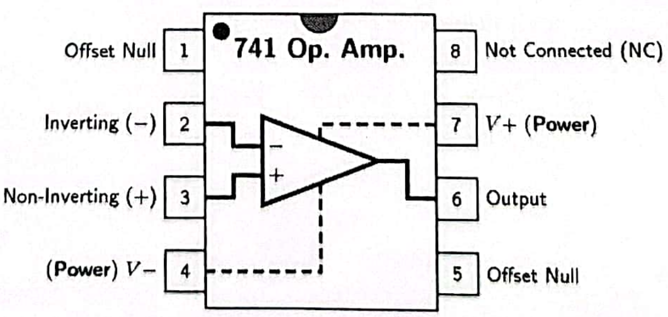
Equipment and component list

Equipment

1. Digital Multimeter
2. DC power supply

Component

- Operational amplifier - UA741 - x1 piece
- Resistors -
  - 10 KΩ - x1 piece
  - 5 KΩ - x1 piece
  - 2.5 KΩ - x1 piece
  - 1.25 KΩ - x1 piece
  - 1 KΩ - x1 piece



741 IC pin diagram

Data Tables

In all the data tables, write the input combinations in ascending order.

SL	$V_D(V)$	$V_C(V)$	$V_B(V)$	$V_A(V)$	$V_Y(V)$
0	0	0	0	0	-0.0035V
1	0	0	0	5	-0.451V
2	0	0	5	0	-0.819V
3	0	0	5	5	-1.267 V
4	0	5	0	0	-1.35V
5	0	5	0	5	-1.80V
6	0	5	5	0	-2.17V
7	0	5	5	5	-2.616V
8	5	0	0	0	-2.25V
9	5	0	0	5	-2.68 V
10	5	0	5	0	-3.07V
11	5	0	5	5	-3.515V
12	5	5	0	0	-3.6 V
13	5	5	0	5	-4.04V
14	5	5	5	0	-4.41V
15	5	5	5	5	-4.86V

Table 1: Table for binary-weighted D/A converter

Signature

Report

Please answer the following questions briefly in the given space.

1. Find the resolution of the D/A converter.

Ans.

$$\Delta = \frac{V_{max} - V_{min}}{2^n} = \frac{-0.0035 - -4.86}{2^4} = 0.304 \text{ V/bit}$$

2. For the D/A converter, change the value of R_F (feedback resistance) to $0.5 \times R_F$ and then to $2 \times R_F$. For each case, measure output voltage for any two consecutive input combinations and calculate the step sizes. Does the effect on step size match with the theory?

Ans.

$0.5 \times R_F$ → Step size = $-2.75 - -2.50 = -0.25V$

$$V_o = \left(\frac{V_A}{R_1} + \frac{V_B \times 2}{R_1} + \frac{V_C \times 4}{R_1} + \frac{V_D \times 8}{R_1}\right)(-0.5 \times R_F)$$

$$V_{o1} = \left(0 + \frac{5 \times 2}{10} + 0 + \frac{5 \times 8}{10}\right)(-0.5 \times 1)$$

$$V_{o1} = -2.5\text{ V}$$

$$V_{o2} = \left(\frac{5}{10} + \frac{5 \times 2}{10} + 0 + \frac{5 \times 8}{10}\right)(-0.5 \times 1)$$

$$V_{o2} = -2.75\text{V}$$

Changing the value of R_F does change the value of step size

$2 \times R_F$

$$V_{o1} = \left(0 + \frac{5 \times 2}{10} + 0 + \frac{5 \times 8}{10}\right)(-2 \times 1) = -10\text{ V}$$

$$V_{o2} = \left(\frac{5}{10} + \frac{5 \times 2}{10} + 0 + \frac{5 \times 8}{10}\right)(-2 \times 1) = -11\text{ V}$$

Step size $= -11 - (-10) = -1\text{V}$

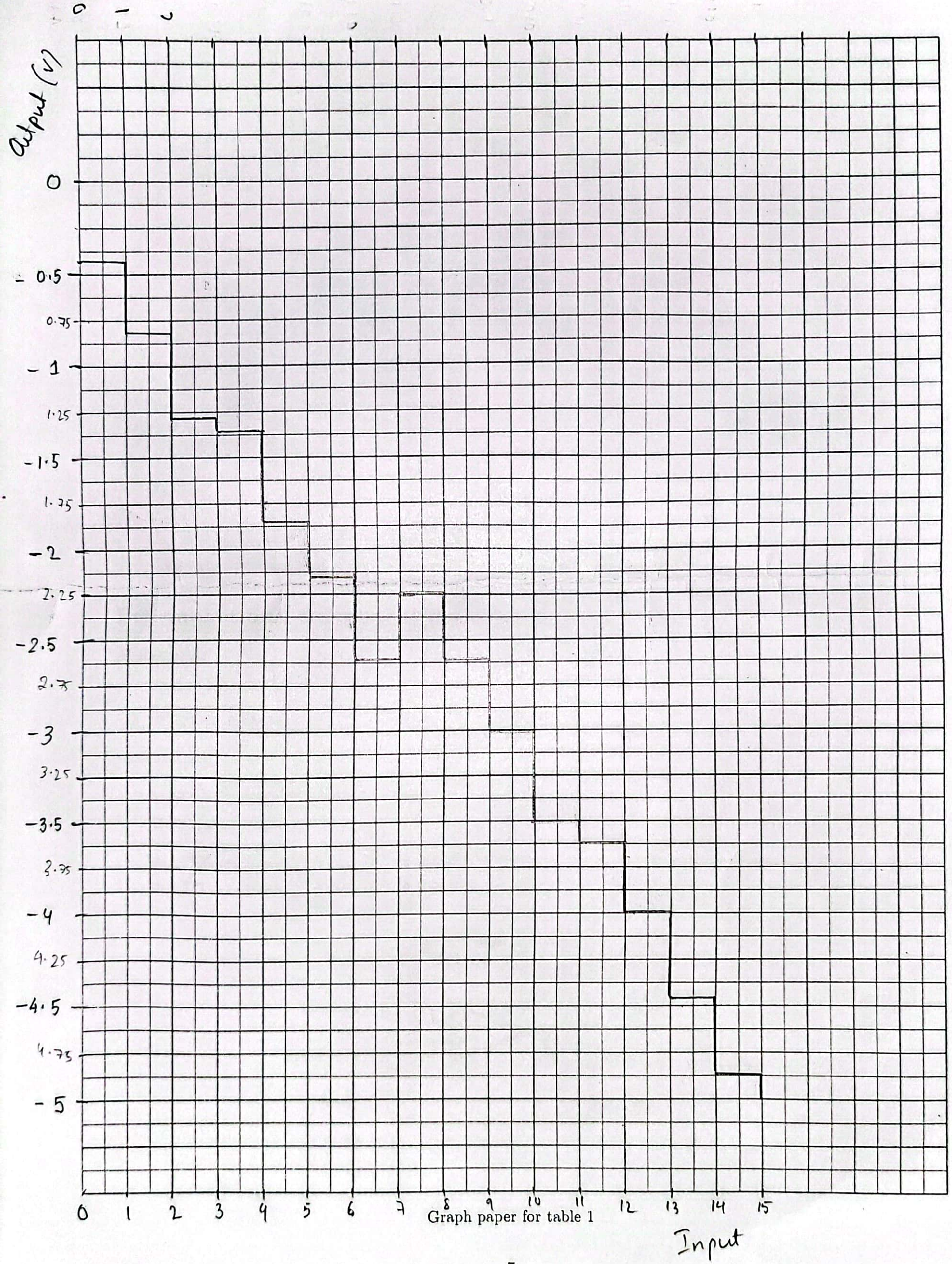
3. How can you get output lower than -15 V in the above D/A converter?

Ans. First, lower the value of $-V_{supply}$.

Increase the value of feedback resistor R_F

or decrease the values of binary wight Resistors (R_1, R_2, R_3, R_4)

4. Plot the results obtained in table 1 in the given graph paper. Keep the serial no of inputs in the horizontal axis and the output voltages in the vertical axis.



Graph paper for table 1